Discrete spectral-triple realization of CFT$_3$-on-S^3 partition function data: a non-holographic alternative to dS/CFT

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Abstract

The boundary CFT$_3$-on-S^3 partition function data of the GeoVac discrete spectral triple — including bit-exact matches to the Klebanov–Pufu–Safdi free energies of the conformally coupled scalar and the free Weyl Camporesi–Higuchi Dirac [12, 18] — is reproduced without any bulk gravitational theory. The framework's S^3 substrate occupies the geometric position that Euclidean dS/CFT [15, 17] assigns to its boundary CFT$_3$, but the construction lacks the bulk dS$_4$ (or its Euclidean partner S^4), the Hartle–Hawking wavefunctional, and the holographic dictionary. We argue that GeoVac sits in a structurally distinct *third category* of correspondence-physics relations, separated from both *Category I* (holographic: AdS/CFT, dS/CFT) and *Category II* (topological: Chern–Simons/WZW). *Category III* is the spectral-triple-discretization category; the framework's analog of the holographic correspondence is the van Suijlekom state-space Gromov–Hausdorff convergence theorem [5] operating between two operator-algebraic discretization levels (discrete spectral triple at finite cutoff $n_{\max}$ and continuum spectral triple on round S^3) at convergence rate $C_3\gamma_{n_{\max}}$. This is not a new holographic correspondence and not a refutation of dS/CFT; it is a structurally distinct realization that produces the same boundary CFT data via a different mechanism. We articulate the taxonomy, place GeoVac within it, and identify the operational consequences of the Category III reading for the framework's existing bit-exact boundary CFT$_3$ results and for the modular structure of the wedge KMS state [7, 8].

Contents

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1 Introduction

1.1 The dS/CFT question

Strominger's dS/CFT proposal [15] posited a duality between gravity on $(d+1)$-dimensional de Sitter spacetime and a Euclidean CFT_d on its conformal boundary. For $d = 3$, the proposal places a CFT_3 on the future timelike infinity I^+ of dS_4, which is geometrically a copy of S^3. Maldacena's Euclidean sharpening [17] via Wick rotation makes the boundary identification precise: Euclidean dS_4 is the round S^4, the equatorial S^3 hosts the boundary CFT_3, and the Hartle–Hawking wavefunctional of dS_4 at I^+ equals the boundary CFT_3 partition function:

$$\Psi_{\mathrm{dS}_4}[\phi_0] \;=\; Z_{\mathrm{CFT}_3}[\phi_0]. \tag{1}$$

The conjectural Eq. (1) relates a 4D bulk quantity (a path integral over geometries with appropriate boundary conditions) to a 3D boundary quantity (a CFT partition function on S^3).

1.2 The GeoVac substrate as boundary geometry

The GeoVac framework [1, 5] constructs a discrete spectral triple $\mathcal{T}_{n_{\max}} = (\mathcal{A}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{\mathrm{GV}})$ on the Fock-projected S^3. This S^3 is the same manifold — same topology, same SO(4) symmetry, same Camporesi–Higuchi spinor bundle [13] — that Eq. (1) assigns as the host of the boundary CFT_3. The framework's discrete spectral-zeta machinery on $\mathcal{T}_{n_{\max}}$ produces bit-exact reproductions of the standard Klebanov–Pufu–Safdi (KPS) [18] closed-form F-theorem coefficients for both the conformally coupled scalar and the free Weyl Dirac [12]. The KPS values are precisely the boundary CFT_3 partition function data that Eq. (1) would predict via Hartle–Hawking.

1.3 The structural question

The GeoVac framework produces the boundary CFT_3 data *without invoking dS_4*: there is no bulk de Sitter geometry in the framework's existing infrastructure, no Hartle–Hawking wavefunctional construction, no functional integral over 4D geometries, and no holographic dictionary mapping bulk fields to boundary operators. The boundary CFT_3 values emerge directly from spectral asymptotics of the discrete spectral triple — specifically, from the spectral-zeta derivative $-\frac{1}{2}\zeta'_\Delta(0)$ evaluated on the round-S^3 continuum limit of D_{GV} [12].

This raises a structural question: *what is the framework's position in the correspondence-physics landscape?* Is GeoVac a non-standard realization of dS/CFT? An alternative to dS/CFT? A different correspondence entirely?

1.4 This paper's contribution

We argue that GeoVac sits in a third structural category, distinct from both AdS/CFT (Category I) and dS/CFT (Category I) on one hand, and from topological constructions like Chern–Simons/WZW (Category II) on the other. Category III is the spectral-triple-discretization category, with two distinguishing features:

1. There is no bulk gravitational theory; the framework's primary object is a discrete spectral triple at finite cutoff, not a 4D geometry to be holographic from.

2. The "correspondence" is between two operator-algebraic objects (the discrete spectral triple at finite cutoff and the continuum spectral triple on round S^3), realized via van Suijlekom state-space Gromov–Hausdorff convergence [5] at rate $C_3\gamma_{n_{\max}} \to 0$.

The Connes–van Suijlekom truncation of the Toeplitz algebra on the unit circle S^1 [14] is the structural precedent for Category III; GeoVac's S^3 construction is the natural non-abelian extension via Paper 38's state-space Gromov–Hausdorff convergence proof (the unconditional lifted-state route; its fifth lemma was withdrawn 2026-09-03 and the theorem does not rest on it).

We do not claim that GeoVac is a new holographic correspondence (it cannot be: there is no bulk). We do not claim it refutes dS/CFT (the boundary data is consistent with what dS/CFT would predict). We argue instead that GeoVac provides a structurally distinct *realization* of the boundary CFT_3-on-S^3 partition function data, via a discrete spectral-triple mechanism that operates differently from holographic reconstruction.

1.5 Outline

Section 2 articulates the three-category taxonomy. Section 3 reviews the GeoVac substrate at the level needed for the structural argument. Section 4 summarizes the bit-exact KPS matches from Paper 50 [12]. Section 5 identifies state-space GH convergence (Paper 38) as the framework's analog of the holographic correspondence. Section 6 addresses the wedge KMS modular structure and its place in the boundary side of correspondence theories. Section 7 explicitly compares the framework to AdS/CFT and dS/CFT. Section 8 lists open questions.

2 The three-category taxonomy of correspondence-physics relations

We organize the landscape of "X/CFT-like" relations into three structural categories.

2.1 Category I — Holographic (bulk gravity ↔ boundary CFT)

Examples. AdS/CFT [16], dS/CFT [15, 17], Kerr/CFT, the flat-space/BMS analog.
Structural ingredients.

- A bulk gravitational theory in $d+1$ dimensions (anti-de Sitter, de Sitter, Kerr, flat space, etc.).

- A boundary theory in d dimensions, typically a CFT.
- A holographic dictionary mapping bulk fields ϕ_{bulk} to boundary operators $\mathcal{O}_{\text{boundary}}$ via an asymptotic relation $\phi_{\text{bulk}}(x, z) \sim z^{\Delta}\mathcal{O}_{\text{boundary}}(x)$ as $z \to 0$.
- A partition-function or wavefunction identification: $Z_{\text{bulk}}[\phi_0] = Z_{\text{boundary}}[\phi_0]$ (AdS/CFT) or $\Psi_{\text{bulk}}[\phi_0] = Z_{\text{boundary}}[\phi_0]$ (dS/CFT).

Operationally. Boundary CFT data is RECONSTRUCTED from bulk gravity by holographic computation. Equivalently, bulk gravity is computed from boundary CFT via reconstruction techniques (RT/HRT minimum surfaces, JLMS bulk-modular reconstruction).

2.2 Category II — Topological (discrete bulk $\leftrightarrow$ boundary state)

Examples. Chern–Simons / WZW boundary [20], state-sum models, tensor networks and MERA.

Structural ingredients.

- A bulk in $d + 1$ dimensions, typically with a discrete or topological structure (lattice, simplicial complex, tensor network).
- A boundary theory on a d-manifold, derived from bulk topological data.
- An identification of the boundary partition function with a bulk topological invariant.

Operationally. Boundary observables emerge as topological invariants of the bulk discrete structure. The bulk is finite or discrete; the boundary CFT (or its analog) emerges in a continuum limit of the discrete bulk data.

2.3 Category III — Spectral-triple discretization (substrate $\leftrightarrow$ continuum CFT)

Examples. Connes–van Suijlekom truncation of the Toeplitz algebra on S^1 [14]; the GeoVac S^3 extension via Paper 38's state-space GH convergence proof [5]; the present paper's KPS-reproducing construction [12].

Structural ingredients.

- A discrete spectral triple $\mathcal{T}_{n_{\max}} = (\mathcal{A}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ at finite cutoff.
- A continuum spectral triple $\mathcal{T}_\infty = (\mathcal{A}_\infty, \mathcal{H}_\infty, D_\infty)$ in the limit.
- A propinquity-style metric Λ on the space of spectral triples.
- Convergence: $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_\infty) \to 0$ at proven rate (Paper 38: $C_3 \cdot \gamma_{n_{\max}}$ with $\gamma_{n_{\max}} \sim (4/\pi) \log n_{\max}/n_{\max}$).
- Continuum CFT$_d$ partition function data is the SPECTRAL content of the limit triple, extracted via zeta-function derivatives (KPS-style).

Operationally. Boundary CFT data is COMPUTED DIRECTLY from spectral asymptotics of the discrete substrate, not reconstructed from any bulk. There is no "bulk" — the discrete substrate is the framework's intrinsic object, and the continuum CFT is what the substrate looks like at the state-space GH limit.

2.4 The three categories at a glance

Category	Bulk object	Correspondence mechanism
I (Holographic)	Bulk gravity in $d+1$ dimensions	Holographic dictionary; reconstruction
II (Topological)	Bulk discrete / topological in $d+1$ dimensions	State-sum; topological invariant
III (Spectral)	Discrete spectral triple at finite cutoff	State-space GH convergence; spectral asy

The three categories are not mutually exclusive in principle: one could imagine constructions that combine, e.g., a holographic bulk with a topological boundary lattice. We do not survey such hybrid constructions here; we focus on the structural placement of GeoVac as a Category III paradigm.

3 The GeoVac substrate

The framework's primary geometric object is the discrete spectral triple

$$\mathcal{T}_{n_{\max}} = (\mathcal{A}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{\mathrm{GV}}) \tag{2}$$

on the Fock-projected [1] round S^3, parameterized by a cutoff $n_{\max} \in \mathbb{N}$. The Hilbert space $\mathcal{H}_{n_{\max}} = \bigoplus_{n=1}^{n_{\max}} \mathcal{H}_n$ is the truncation of the full L^2 spinor bundle on S^3 at the $n_{\max}$-th shell; the algebra $\mathcal{A}_{n_{\max}}$ is the (algebraic) Fock multiplier algebra at that cutoff; D_{GV} is the discrete Camporesi–Higuchi Dirac operator with eigenvalues $|\lambda_n| = n + 3/2$ and multiplicities $g_n^{\mathrm{Dirac}} = 2(n+1)(n+2)$ on the spinor bundle.

The continuum limit $\mathcal{T}_\infty$ is the round-S^3 Camporesi–Higuchi spectral triple on the full spinor L^2, with the same eigenvalue structure extended to all $n \in \mathbb{N}$.

The substrate has no "higher-dimensional" geometry. There is no 4D bulk, no internal z-coordinate, no holographic radial direction. The S^3 is the framework's natural geometric arena, intrinsic to the Fock projection of the Coulomb spectrum [1].

Remark 3.1 (Why no bulk)**.** A natural question is whether the GeoVac framework could be *extended* with a 4D bulk geometry (Lorentzian dS_4 or Euclidean S^4). The substrate's existing infrastructure does not contain any such geometry: Paper 50 [12] §6 documents that the framework's codebase has no AdS_4 / H^4 / dS_4 / S^4 machinery, and Sprint RH-B[1] (2026-04-17) [3] closed the $S^3 \to H^3$ Wick continuation as a clean structural dead end (the continuation supplies no natural discrete subgroup $\Gamma \subset \mathrm{SO}(3,1)$ selectable from framework invariants). Importing a 4D bulk would be a major structural extension, beyond sprint scale, outside the present paper's scope and would, in any case, be an *extension* of the framework rather than something already present.

4 The boundary CFT_3 partition function data

Paper 50 [12] establishes that the GeoVac substrate reproduces standard boundary CFT_3-on-S^3 partition function data bit-exactly. We summarize the headline results here for self-containment.

4.1 Conformally coupled scalar

Theorem 4.1 (Paper 50 [12], Theorem 3.1)**.** *On the truncated GeoVac spectral triple $\mathcal{T}_{n_{\max}}$, the spectral-zeta derivative of the conformally coupled scalar Laplacian on round unit S^3 satisfies*

$$F_s^{S^3} \;:=\; -\tfrac{1}{2}\,\zeta'_{\Delta_{\mathrm{conf}}}(0) \;=\; \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2}. \tag{3}$$

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

This matches the Klebanov–Pufu–Safdi closed form [18] at 61+ *digits and admits the PSLQ integer relation* $[8, 2, -3]$ *across the basis* $\{\log 2, \zeta(3)/\pi^2\}$.

4.2 Free Weyl Dirac

Theorem 4.2 (Paper 50 [12], Theorem 3.2)**.** *On the truncated GeoVac spectral triple* $\mathcal{T}_{n_{\max}}$*, the spectral-zeta derivative of the free Weyl Camporesi–Higuchi Dirac on round unit* S^3 *satisfies*

$$F_D^{S^3} \;:=\; -\tfrac{1}{2}\,\zeta'_{D_{\mathrm{CH}}}(0) \;=\; \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2}. \tag{4}$$

This matches the KPS closed form symbolically (sympy-exact).

4.3 Orthogonal master Mellin engine decomposition

The scalar and Dirac F-values combine linearly to extract orthogonal master Mellin engine sectors [2, 12]:

$$F_D + 2F_s \;=\; \frac{\log 2}{2} \qquad \text{(M2 only, all } \zeta(3) \text{ cancels)} \tag{5}$$

$$F_D - 2F_s \;=\; \frac{3\zeta(3)}{4\pi^2} \qquad \text{(M3 only, all } \log 2 \text{ cancels).} \tag{6}$$

This is the framework's first verification of the master Mellin engine on non-GeoVac-internal physics observables [4, 12].

4.4 The data the framework reproduces

Eqs. (3)–(6) are standard CFT_3 data. Eq. (3) is the F-theorem coefficient [18] that monotonically decreases along renormalization-group flows of 3D CFTs. The Maldacena identification (1) would predict exactly these values via the holographic computation of the Hartle–Hawking wavefunctional of dS_4 at $\mathcal{I}^+$. The framework reproduces them without that computation.

5 State-space GH convergence as the "correspondence" mechanism

The framework's analog of the holographic correspondence is van Suijlekom state-space Gromov–Hausdorff convergence between the discrete spectral triple at finite cutoff and its continuum limit.

5.1 Statement of the convergence theorem

Theorem 5.1 (Paper 38 [5]; WH1 PROVEN)**.** *The discrete GeoVac spectral triples* $\mathcal{T}_{n_{\max}}$ *on the Fock-projected* S^3 *converge to the continuum round-*S^3 *Camporesi–Higuchi spectral triple* $\mathcal{T}_\infty$ *in the van Suijlekom state-space Gromov–Hausdorff sense:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_\infty) \;\le\; C_3 \cdot \gamma_{n_{\max}} \;\to\; 0 \quad \textit{as } n_{\max} \to \infty, \tag{7}$$

where $C_3 = 1$ *asymptotically (Lipschitz comparison constant) and* $\gamma_{n_{\max}} = O(\log n_{\max}/n_{\max})$ *is the central-Fejér moment with universal asymptotic rate constant* $4/\pi$.

The proof is the lemma chain L1′-L2-L3-L4 of Paper 38 [5] together with its lifted-state construction, generalized to all simple compact Lie groups in Paper 40 [6] (for general G conditional on a per-group spin-window decomposition, verified for SU(2)). Paper 38's fifth lemma — the Latrémolière propinquity assembly — was *withdrawn* on 2026-09-03 when its height leg was refuted; the convergence theorem is unaffected, because the unconditional route uses only reach-type estimates and no height constituent.

5.2 Why this is the "correspondence" mechanism

The state-space GH convergence relates two operator-algebraic objects (the discrete and continuum spectral triples) by a quantitative metric, with proven convergence rate. The boundary CFT_3 partition function data of Section 4 is extracted from the continuum spectral triple $\mathcal{T}_\infty$ via the zeta-function derivative $-\frac{1}{2}\zeta'(0)$; this derivative is PRESERVED by the state-space GH limit because

- the spectral data of $D_{n_{\max}}$ converges to that of D_∞ in the operator-norm sense (per Berezin reconstruction, Paper 38 L4);
- the spectral-zeta function converges term-by-term and uniformly on the strip $\mathrm{Re}(s) > d/2$;
- the meromorphic continuation to $s = 0$ commutes with the state-space GH limit by standard analytic-continuation arguments.

In holographic terminology, this is the framework's "dictionary": the discrete spectral triple at finite cutoff is the framework's "bulk-like" object (more precisely, its substrate), and the boundary CFT_3 data is what that object looks like at the state-space GH limit. There is no separate "bulk theory" to translate from; the spectral triple at finite cutoff is itself the framework's primary object.

5.3 Difference from holographic reconstruction

Holographic reconstruction recovers boundary CFT data from bulk gravity by integrating field-theoretic correlators against bulk modes. The boundary CFT exists in the same dimensional setting as the bulk's asymptotic boundary, and the dictionary translates between two distinct physical theories.

State-space GH convergence recovers continuum CFT data from a discrete substrate by taking the continuum limit of the substrate's spectral asymptotics. Both objects (discrete and continuum spectral triples) live in the same dimensional setting (both are spectral triples on S^3); the dictionary translates between two operator-algebraic discretization levels, not between two physical theories of different dimension.

This is a structurally distinct mechanism. It is appropriate to call both a "correspondence" in the abstract sense (each relates mathematical structures in a way that yields physical data), but the specific nature of the correspondence differs.

6 Wedge KMS modular structure

A second structural feature of the GeoVac framework that overlaps with holographic theories is the wedge KMS modular structure [7, 8, 12].

6.1 Wedge KMS state on the framework

Paper 50 [12] defines a wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ on a hemispheric wedge of the S^3 substrate, where K_α^W is the modular Hamiltonian associated with the boost generator of the wedge. The state ρ_W has the structural properties of the Bisognano–Wichmann vacuum [19].

Paper 42 [7] establishes the four-witness Wick-rotation theorem: the modular structure of ρ_W corresponds bit-exactly with the thermal-state readings of Hawking, Sewell, Bisognano–Wichmann, and Unruh at finite cutoff $n_{\max}$. The Tomita–Takesaki modular flow on the wedge KMS state is a literal identification at the operator-system level (Riemannian).

6.2 Connection to holographic theories

Any holographic correspondence involving thermal states (AdS-Schwarzschild $\leftrightarrow$ thermal CFT, dS-Schwarzschild $\leftrightarrow$ static patch CFT, BTZ $\leftrightarrow$ thermal CFT on $S^1 \times \mathbb{R}$) has the SAME modular-flow structure on its boundary CFT side. The four-witness theorem of Paper 42 explicitly identifies Hawking, Sewell, Bisognano–Wichmann, and Unruh as four readings of one modular-flow structure.

GeoVac therefore shares the boundary-modular-flow content with all holographic-correspondence theories. The Connes–Rovelli thermal-time identification of Paper 49 [11] makes this concrete: thermal time is the modular flow, regardless of whether one came at it from a bulk-gravity calculation or a direct boundary-CFT analysis.

The structural distinction we are drawing is therefore:

- The boundary-modular-flow content is SHARED across Category I and Category III.
- The bulk content distinguishes them: Category I has a bulk gravitational theory; Category III has a discrete spectral-triple substrate.

7 Comparison to AdS/CFT and dS/CFT

We summarize the comparison in a cross-table.

Theory	Bulk object	Correspondence mechanism
AdS_5/CFT_4 [16]	AdS_5 IIB SUGRA	Holographic dictionary
AdS_4/CFT_3 (ABJM)	AdS_4 M-theory	Holographic dictionary
dS_4/CFT_3 [15, 17]	dS_4	Hartle–Hawking wavefunctional
Kerr/CFT	Kerr	Asymptotic-symmetry holography
Chern–Simons/WZW [20]	3D topological	State-sum
Connes–vS S^1 [14]	Discrete spectral triple on S^1	Propinquity (Toeplitz S^1)
GeoVac (Papers [5, 12])	**Discrete spectral triple on S^3**	**State-space GH convergence at rate 4/**

7.1 What GeoVac shares with dS/CFT

- The same boundary host manifold: the Euclidean dS/CFT identification (1) places the boundary CFT_3 on S^3; GeoVac's substrate is on the same S^3.
- The same boundary CFT_3 partition function data: Eqs. (3), (4) reproduce values that Hartle–Hawking would predict.
- The same wedge KMS / modular-flow structure on the boundary side (four-witness theorem, Paper 42 [7]).

7.2 What GeoVac does not share with dS/CFT

- GeoVac has no bulk dS_4 (Lorentzian) or S^4 (Euclidean).
- GeoVac has no Hartle–Hawking wavefunctional construction.
- GeoVac has no functional integral over 4D geometries.
- GeoVac has no holographic dictionary mapping bulk fields to boundary operators.

7.3 The Category III alternative

The framework's mechanism for producing the boundary CFT_3 data is the state-space GH convergence of Theorem 5.1: the discrete spectral triple at finite cutoff approaches the continuum round-S^3 spectral triple in an explicit metric, with proven convergence rate. The boundary CFT_3 data is the spectral-zeta content of the continuum limit, and this content is preserved by the state-space GH limit by standard analytic-continuation arguments. This is structurally distinct from any holographic mechanism.

Remark 7.1 (What this paper is NOT claiming)**.** We explicitly do not claim:

1. GeoVac is a new holographic correspondence (no bulk = no holography in the standard sense).
2. GeoVac is a refutation of dS/CFT (the boundary data is consistent; we just produce it differently).
3. GeoVac is a Theory of Everything (per the rhetoric rule of the GeoVac project [1], the framework does not assert ontological priority of the spectral-triple description over continuum quantum mechanics).
4. GeoVac proves Maldacena's identification (1) (we have no access to the bulk side).

What we do claim is that GeoVac produces the same boundary CFT_3 data via a structurally distinct mechanism that places the framework in a third category, distinct from holographic correspondences.

8 Open questions

Q1. The framework's bulk-side identifications corresponding to Hartle–Hawking dS_4 (or its Euclidean S^4 partner), if any, remain unreachable from the existing infrastructure. Importing 4D bulk machinery would be a major structural extension, beyond sprint scale. Whether such an extension is structurally compatible with the framework's existing Coulomb / Fock-projection origin is open.

Q2. The Connes–van Suijlekom Toeplitz S^1 truncation [14] is the abelian Category III precedent. Paper 40 [6] establishes the universal rate constant $4/\pi$ for all simple compact Lie groups; the state-space Gromov–Hausdorff convergence itself is rigorous for SU(2) and, for general G, conditional on a per-group spin-window decomposition (a named gap). Whether the boundary CFT data for other groups (e.g., SU(3), Sp(2), G_2) similarly reproduces standard CFT-on-group partition functions is a natural extension.

Q3. The boundary CFT_3 data of Eqs. (3), (4) reproduces free-field KPS values. Whether the framework extends to interacting CFT_3 partition function data is open. The master Mellin engine [2, 4] suggests that the transcendental signature of any CFT_3 partition function decomposes under M1/M2/M3, but mapping this decomposition to interacting CFT_3 data requires further analysis.

Q4. A metric-level Lorentzian extension of Category III to non-compact carriers does *not* currently close: Papers 45 [9] and 46 are descoped (the K^+ compression annihilates the Lipschitz seminorm — a degeneracy theorem, not a Lorentzian convergence), and only Paper 47 [10]'s norm-resolvent arrow survives (partial). Whether any future Lorentzian extension produces dS-thermal-state observables (Hawking radiation on the discrete substrate, thermal density of states, etc.) that match dS/CFT predictions is open.

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